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Crack-branching

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ABSTRACT

Velocity measurements on cracks in sapphire and glass have been made using Wallner lines. Contrary to the view that crack-branching in a given material occurs at a constant critical velocity, it is found that the branching velocity can vary with the stress on the crack. On the other hand, $\sigma_f c_b^{\frac{1}{2}}$ is constant. Here σ_f is the stress normal to the crack and c_b is the semi-length of a double-ended crack when it branches. A theory is given in terms of the condition for inducing persistent growth of a new crack ahead of the running crack.

§ 1. INTRODUCTION

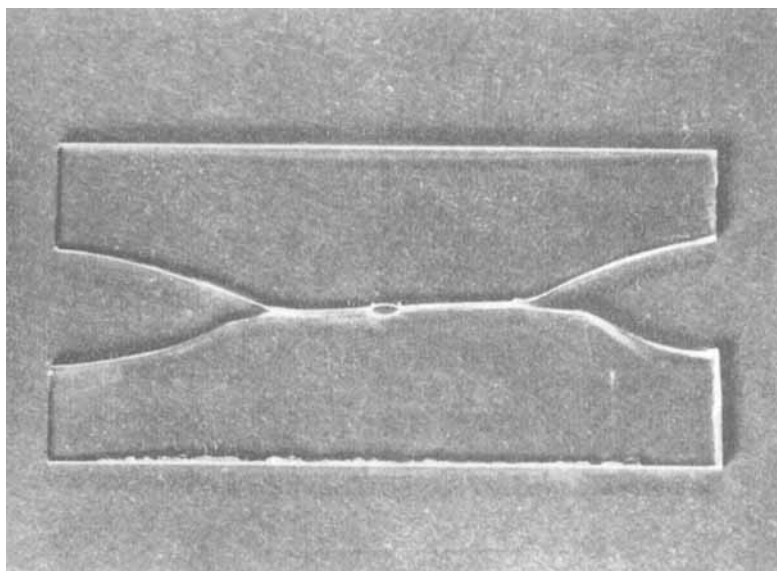
WHEN a brittle crack extends under constant stress, it accelerates and eventually branches (fig. 1).

The stress distribution ahead of a crack changes as the crack velocity increases. Yoffe (1951) first showed that above about 0.6 of c_2 , the elastic shear wave velocity, the normal stress on two symmetrically inclined planes through the crack tip becomes greater than that on the crack plane itself. Yoffe considered a crack of constant length moving at constant velocity under a given applied stress. For an extending crack ahead of a moving wedge, Barenblatt and Cherepanov (1960) obtained a similar result. The value of the critical velocity for the stress change depends somewhat on Poisson's ratio. Because of these calculations it is usual to associate crack-branching with the stress distribution effect and to suppose that branching occurs at a critical velocity.

The higher the stress on the crack, the shorter the distance the crack travels before the branch forms. This could reflect faster acceleration to the critical velocity. However, measurements on the relationship between the applied stress and the crack size at branching have recently been used in conjunction with Berry's (1960) crack-velocity calculation to determine the effective surface energy of a running crack in alumina, magnesia and glass (Congleton and Petch 1965) and part of this analysis leads to the conclusion that branching does not occur at a constant velocity. The higher the stress on the crack, the lower the branching velocity.

The initial purpose of the present work was to check this conclusion by direct measurement.

Fig. 1



Branching of a crack growing from a central hole in a glass plate.

§ 2. VELOCITY MEASUREMENTS

The velocity of a crack can be determined from certain fracture-surface markings using a technique developed by Wallner (1939), Smekal (1950) and Shand (1954). These markings (Wallner lines) arise from the interaction of an elastic shear wave with the crack front and it is possible to measure crack velocities using the shear wave as the timing device. The method has been described in the above references and only brief details of the procedure used in the present measurements need be given here.

Let the fracture origin be at O on the fracture surface (fig. 2(a)) and let a shear wave be initiated due to some disturbance when the crack front reaches S, then SBS' is the Wallner line that arises by interaction between the crack front spreading from O and the shear wave spreading from S.

The average crack velocity over a distance interval such as EF can now be calculated. When the crack front is at E, it is also at E' if the crack spreads with a uniform radial velocity to give a circular front centred on O. Similarly, when the crack front is at F, it is also at F'. Since E' and F' are points on the Wallner line, they are also points on the shear wave front spreading from S. Thus, provided the shear wave has a uniform radial velocity:

$$\text{average crack velocity over EF} = \frac{EF}{(SF' - SE')} \cdot c_2.$$

This type of measurement can be repeated with each Wallner line on the surface.

In the present experiments, another set of measurements was also possible that has not been described before. Since thin plates were used, the shear waves were reflected from the faces of the plate and interacted again with the crack front, creating secondary Wallner lines ($S'C$ in fig. 2 (b)). In the distance interval B to C, the crack front moves from B to C while the shear wave moves from B to S' and is reflected back to C. Thus:

$$\text{average crack velocity over BC} = \frac{BC}{(BS' + S'C)} \cdot c_2.$$

Fig. 2

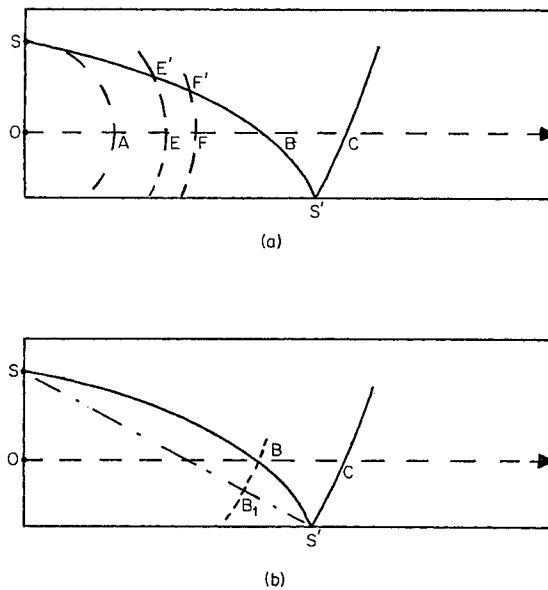


Diagram of primary and secondary Wallner lines on a fracture surface.

Since B and C are both points on Wallner lines, this calculation is independent of the crack front shape. Strictly, BS' should be replaced by B_1S' , where BB_1 is part of a circle centred on S. In practice, this makes little difference.

In the present work these procedures were applied to Wallner lines on annealed soda lime glass and sapphire fracture surfaces obtained in plates ($50 \times 30 \times 1$ mm and $40 \times 20 \times 1$ mm respectively) broken from central 2.3 mm long cracks formed by ultrasonic drilling. The method of applying a uniaxial tension normal to the crack for the fracture of these plates has already been described in the surface energy measurements (Congleton

Fig. 3

Wallner lines on a glass fracture surface $\times 40$.

and Petch 1965). The sapphire plates had a (1010) plane as the 4×2 cm face and the (0001) plane made an angle of 14° with the 4 cm edge. The tension was applied parallel to this edge.

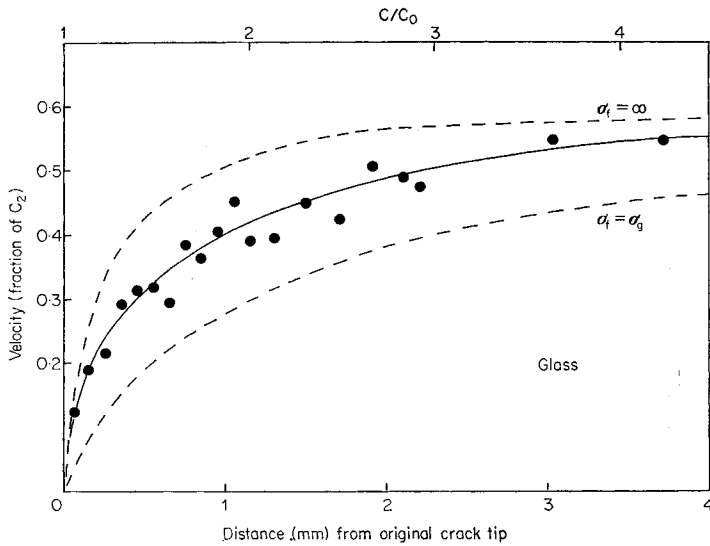
The specimens were fractured at a range of stresses obtained by some variation of the crack tip radius in the range of 10^{-2} – 10^{-3} cm and by variation of the temperature and environment (table 1). The Wallner line measurements were made on photographs of the fracture surface at $\times 50$ (fig. 3).

Table 1. Fracture data in the velocity measurements

Material	Fracture stress 10^8 dyne cm^{-2}	Expt. conditions
Sapphire	9.08	Liqd. N_2 , -196°C
"	5.48	Transformer oil, 18°C
"	4.59	Transformer oil, 18°C
"	4.72	Water, 18°C
Glass	3.47	Liqd. N_2 , -196°C
"	2.84	Liqd. N_2 , -196°C
"	3.00	Silicone oil, 18°C
"	1.75	Acetone + sold. CO_2 -78°C

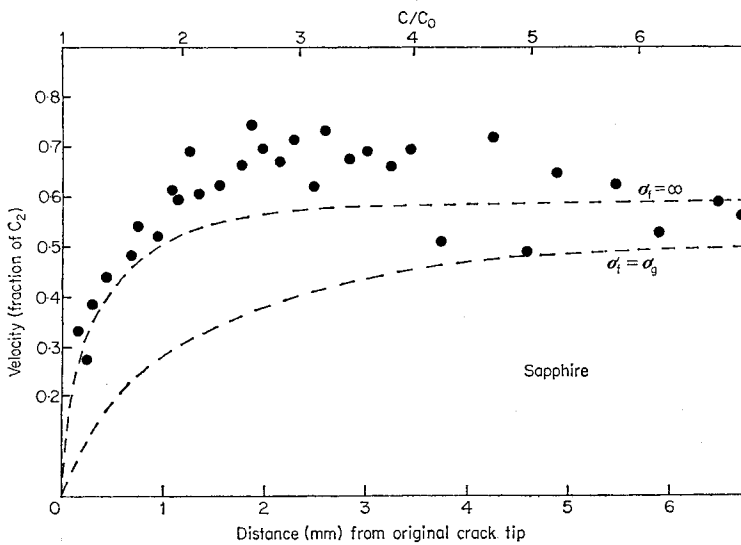
The measured crack velocities are shown in figs. 4 and 5. Within the accuracy of the experiments, no certain difference was detected between the results at the different stresses, so the average curves are plotted. Using these curves along with measurements of the crack size at branching, fig. 6 was obtained. This gives the relationship between the stress on the crack and its velocity at branching. Also included in this figure are the branching velocities deduced in the course of the previous surface energy measurements.

Fig. 4



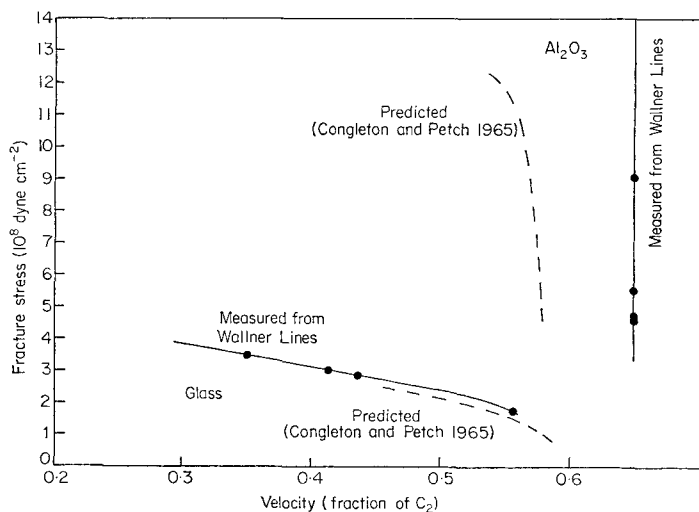
Measured velocity of a crack in glass and the theoretical limit curves (c_0 of original crack = 1.15 mm).

Fig. 5



Measured velocity of a crack in sapphire and the theoretical limit curves (c_0 of original crack = 1.15 mm).

Fig. 6



Variation of the velocity at branching with the fracture stress for glass and sapphire (c_0 of original cracks = 1.15 mm).

§ 3. DISCUSSION OF THE VELOCITY MEASUREMENTS

According to Berry's (1960) extension of Mott's (1948) calculation, the velocity v of a symmetrical crack that has grown to a half-length c is:

$$v = v_m \left\{ 1 - \left(\frac{c_0}{c} \right)^2 - 2 \left(\frac{\sigma_g}{\sigma_f} \right)^2 \frac{c_0}{c} \left(1 - \frac{c_0}{c} \right) \right\}^{1/2},$$

where v_m is the maximum velocity, c_0 is the initial half-length, σ_f is the actual fracture stress and σ_g is the fracture stress given by the Griffith criterion for a c_0 crack.

There are two limiting cases:

$$\begin{aligned} v &= v_m \left(1 - \frac{c_0}{c} \right); & \sigma_f &= \sigma_g, \\ v &= v_m \left(1 - \left(\frac{c_0}{c} \right)^2 \right)^{1/2}; & \sigma_f &= \infty; \end{aligned}$$

these are also plotted in figs. 4 and 5 taking v_m from Roberts and Wells (1954) as $0.6 c_2$ for an isotropic solid with Poisson's ratio of $\frac{1}{4}$.

In the case of glass (fig. 4), the measured velocities lie between the two theoretical limits and tend towards $0.6 c_2$ in agreement with Roberts and Wells.

With sapphire (fig. 5) the initial crack velocities measured from the primary Wallner lines rise above the upper theoretical limit, but the later values, measured from the secondary Wallner lines, are within the limits. Sapphire is not elastically isotropic, so c_2 will have some dependence on direction; also, because of crystallographic influences, the

crack front is probably not quite circular. Thus, the velocity measurements for sapphire are not so reliable as those for glass.

In fig. 6 the directly measured branching velocities, particularly for glass, are in good agreement with those previously deduced in the surface energy measurements using the Berry equation with $v_m = 0.6 c_2$.

It is quite clear from fig. 6 that branching does not occur at a constant velocity. With glass there is a considerable range of branching velocity. With sapphire, branching did not occur in the present measurements until the crack had nearly reached its maximum velocity; thus, although the crack size at branching varied with the fracture stress, this did not represent much variation in velocity.

It is also apparent from fig. 6 that branching can occur before the velocity is reached at which Yoffe's calculation indicates that the stress normal to an inclined plane through the crack tip is greater than the stress normal to the crack plane. However, the velocities are sufficiently high that the stress should be fairly constant over an appreciable arc about the head of the crack.

§ 4. THE CRACK-BRANCHING CRITERION

Since crack-branching does not occur at a constant velocity, some other factor must be important. The view adopted in this paper is that the prime factor in branching is the growth of new cracks ahead of the initial running crack. The constancy of stress over an arc because of the velocity may be a contributory factor that facilitates the development of the new cracks at an inclination to the original.

It seems probable that the condition for the growth of these new cracks should be a critical stress field ahead of the running crack. In the terminology of fracture mechanics (Irwin 1957), the criterion for crack-branching should then be a critical stress intensity factor K near the tip of the running crack.

Thus, branching should occur when

$$K = \pi^{1/2} \sigma_f c_b^{1/2} = \text{constant}, \quad . \quad . \quad . \quad . \quad . \quad (1)$$

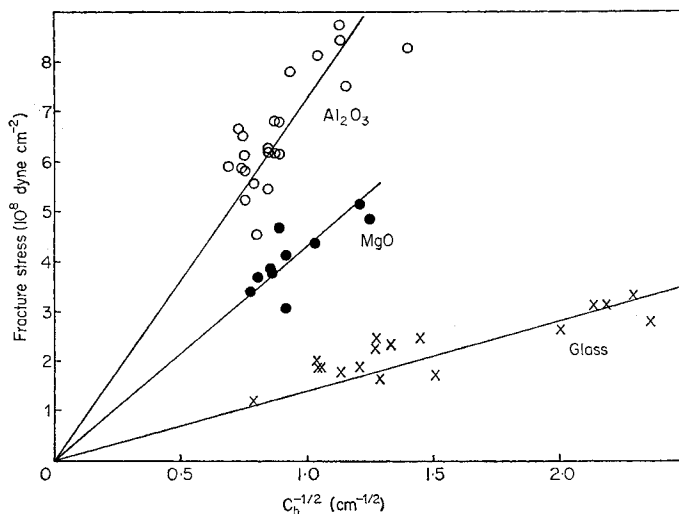
where c_b is the semi-length of the crack at branching.

Figure 7 gives the results of branching measurements on $50 \times 30 \times 1$ mm plates of polycrystalline alumina (Lucalox) polycrystalline magnesia and soda lime glass broken in the uniaxial tension apparatus from central 2.3 mm long cracks. The experiments were performed at room temperature in silicone oil, which nearly eliminates any static fatigue effects.

It is apparent that eqn. (1) is obeyed.

Measurements by Levengood (1958), Shand (1959) and Kerper and Scuderi (1964) on the length of the smooth 'mirror' on a glass fracture surface give further support to eqn. (1), since the mirror ends just before branching occurs.

Fig. 7



Fracture stress versus $c_b^{-1/2}$ for glass and for polycrystalline alumina and magnesia. Measurements in silicone oil at room temperature.

Very recently, Clark and Irwin (1966) have also concluded that branching is by advance cracking and requires a critical stress intensity factor. However, they believe the branching does not occur until the velocity closely approaches the limiting value.

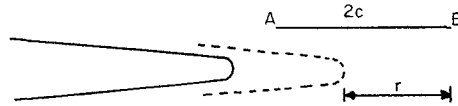
Table 2 gives the $\sigma_f c_b^{1/2}$ values obtained from fig. 7. Values are also included from measurements at -196°C (temperature too low for static fatigue), at room temperature in water (static fatigue favoured) and at 100°C in silicone oil (protected from static fatigue). In these measurements, the c_b values were sufficiently small relative to the plate width that departure from the infinite width stress distribution (Dixon 1959) could have little effect on $\sigma_f c_b^{1/2}$.

Table 2. $\sigma_f c_b^{1/2}$, $10^8 \text{ dyne cm}^{-3/2}$

Condition	Alumina	Magnesia	Annealed glass	As-received glass
Room temp., silicone oil	7.3	4.3	1.2	1.6
-196°C	9.2	5.4	1.5	1.6
Room temp., water.	6.5	4.6	—	1.2
100°C silicone oil	6.9	3.6	—	1.4

To examine the basis for (1) more closely, consider an adventitious Griffith crack of length $2c$ ahead of the main crack, fig. 8. The closer the two are together, the greater the stress on the Griffith crack. However, the main crack cannot get nearer than c from the centre of the Griffith crack without either absorbing it or beginning to pass. Once passing begins, the stress falls at the A end, although initially it continues to rise at B.

Fig. 8



Model for crack-branching calculation.

Using the static stress solution (e.g. Westergaard 1939), which should not lead to much error at the velocities involved, the stress normal to the crack plane at a distance c ahead of the crack tip at branching is $\sigma_f(c_b/2c)^{1/2}$, provided $c \ll c_b$.

Thus, the condition for extension of the $2c$ Griffith crack by this stress when the centre of the Griffith crack is c ahead of the main crack is approximately:

$$\left. \begin{aligned} \sigma_f \left(\frac{c_b}{2c} \right)^{1/2} &= \left(\frac{2E\gamma}{\pi c} \right)^{1/2}, \\ \sigma_f c_b^{1/2} &= 2 \left(\frac{E\gamma}{\pi} \right)^{1/2}, \end{aligned} \right\} \dots \dots \dots (2)$$

where E is Young's modulus, γ is the effective surface energy and plane stress is considered.

Thus, (2) should give fairly closely the minimum size of the main crack that can activate extension of a fresh Griffith crack ahead, when σ_f is the applied stress. This suggests that (2) may be the criterion for branching.

Table 3 gives the experimental evidence on this. Since the specific surface energy is $\sim (Ea/30)$, where a is the interatomic spacing, $\sigma_f c_b^{1/2}$ should be approximately proportional to E if the criterion for branching is given by (2). Table 3 shows experimental agreement with this. However, the calculated values of $\sigma_f c_b^{1/2}$ in the table, obtained using measured values of the specific surface energy in (2), are low.

Table 3. Values of $\sigma_f c_b^{1/2}$ (dyne cm^{-3/2})

Material	$\sigma_f c_b^{1/2}$ (obs., fig. 7)	E (dyne cm ⁻²)	$\sigma_f c_b^{1/2}$ obs/ E (cm ^{1/2})	γ (erg cm ⁻²)	$\sigma_f c_b^{1/2}$ (calc.)
Al ₂ O ₃	7.3×10^8	4.0×10^{12}	1.8×10^{-4}	2000	1×10^8
MgO	4.3×10^8	3.0×10^{12}	1.4×10^{-4}	1000	6×10^7
Glass	1.6×10^8	6.8×10^{11}	2.3×10^{-4}	500	2×10^7

For two main reasons these low calculated values of $\sigma_f c_b^{1/2}$ can be expected. First, the use of the specific surface energy under-estimates the stress required for extension of a crack. The effective surface energy for crack extension seems to be about 5–25 times the specific surface energy for the materials used here (Shand 1961, Clarke *et al.* 1966, Congleton and Petch 1965, 1966). This raises the calculated $\sigma_f c_b^{1/2}$ by 2–5 times.

The effective γ for crack extension increases as the temperature is lowered and decreases if the environment permits static fatigue (Congleton and Petch 1966). Thus, the high strain rate ahead of a fast crack should favour a high value of the effective γ in crack branching and the factor on the calculated $\sigma_f c_b^{1/2}$ is likely to be at the upper end of the 2–5 range. The values of observed $\sigma_f c_b^{1/2}$ in table 2 suggest that some influence of temperature and possibly of environment on the effective γ persists even at the high strain rate in the branching process.

A second reason for expecting low calculated values of $\sigma_f c_b^{1/2}$ in table 3 is the insufficiency of merely initiating Griffith crack extension ahead of the main crack; this extension must persist if branching is to be achieved. The problem is a dynamic one. If immediately after extension of the Griffith crack begins, the main crack front sweeps past, the Griffith crack will be unloaded and its growth arrested. Thus, the initial extension of the Griffith crack must begin sufficiently far in advance of the main crack that the two are moving at comparable velocities when the main one catches up.

Berry's equation (fig. 4) indicates that when the half-length of the Griffith crack has extended by $\sim 2c$, its velocity will be comparable to that of the main crack at branching. Over this distance, the average velocity of the Griffith crack is about half that of the main crack. Thus, if extension of the Griffith crack begins when its centre is c ahead of the main crack tip, the tip will catch up with the far end of the Griffith crack. This will be avoided if the Griffith crack is a little further ahead ($\sim 2c$) when extension begins; the stress on the Griffith crack should then remain sufficiently high to maintain extension. This change in the position of the Griffith crack from the one used in the deduction of (2) increases the calculated values of $\sigma_f c_b^{1/2}$ in table 3, but only by a factor of $\sqrt{2}$.

Even when unloading of the Griffith crack is avoided in this way, the establishment of branching may still be a little uncertain. The main crack can be arrested locally by running into the Griffith crack, but the stress ahead of the latter will then be sufficient to induce a second advance crack. Thus, some rather complex cracking can be expected at this stage. The situation will stabilize once two cracks are established growing away from each other. Because of competition between the two, the stress intensity factor at each will be insufficient to induce further branching until more crack growth has occurred.

The formation of new cracks in advance of the main crack appears to be in keeping with the appearance of fracture surfaces in very brittle solids. Typically, the initial growth of the fracture gives a smooth mirror appear-

ance, then there is a zone of increasing roughness, containing numerous small subsidiary cracks, before branching occurs. The roughness could correspond to Griffith cracks that begin to extend but are unable to establish branching.

In table 3, once the two modifications discussed above (the use of effective surface energies and the 'persistent growth' consideration) are introduced into the calculation of $\sigma_f c_b^{1/2}$ using (2), good agreement is obtained between calculated and observed values.

It is concluded that the approximate criterion for branching is that the stress at $\sim 2c$ ahead of the main crack must be able to induce persistent extension of a $2c$ Griffith crack located there.

The above discussion is in terms of pre-existing Griffith cracks and this is probably appropriate in the present case, since there was no special surface treatment of these brittle specimens. In other cases, e.g. crack-branching in strong metals or crack-free brittle solids, it is to be expected that the Griffith crack has to be generated by plastic deformation at the main crack. Provided the critical event remains the propagation of the Griffith crack, the essence of the present treatment should stand, but if the details of the crack generation become critical, the problem will be more involved.

It seems probable that the study of crack-branching in strong metals may make a useful contribution to understanding their fracture toughness properties.

§ 5. AN ENERGY CRITERION FOR BRANCHING

Since this work was completed, Johnson and Holloway (1966) have discussed a number of interesting aspects of branching and they have calculated a branching criterion purely by an energy argument. This considers the supply of sufficient energy to form four, instead of two, new surfaces at each end of the main crack, i.e. to branch.

Once a crack begins to propagate at constant stress, the rate of supply of driving energy exceeds the rate of its consumption as surface, and strain energy and the excess appears as kinetic energy. When the rate of supply becomes sufficiently large, it should be possible to support formation of a branch crack and the additional surface energy required can be obtained at the expense of the contribution to kinetic energy. Johnson and Holloway assume that during an increment in crack length at branching a proportion of the driving energy supplied by the applied stress must still go into additional kinetic energy. But there seems no reason for this; the whole of the increment of driving energy can be converted into strain, and surface energy. The new length of crack will acquire kinetic energy, but this can be obtained from the kinetic energy associated with the existing length of crack, i.e. the crack velocity drops.

With this modification, branching is energetically possible when

$$\sigma_f = 2 \left(\frac{E\gamma}{\pi c_b} \right)^{1/2}, \quad . \quad . \quad . \quad . \quad . \quad (3)$$

i.e. when the Griffith equation is satisfied for a doubled surface energy.

The energy criterion for branching could in principle be fulfilled at a somewhat smaller c_b than given in (3) by drawing on the existing kinetic energy of the crack to supply surface energy.

There is always some uncertainty about energy arguments. They give criteria that are necessary but that may not be sufficient, since they do not consider whether there is a practicable mechanism by which the process in question can be achieved. However, it will be seen that the energy criterion in (3) is in fact the same as the stress criterion obtained in (2) for initiation of a branching mechanism.

§ 6. CONCLUSIONS

1. Branching does not occur at a constant crack velocity, but $\sigma_f c_b^{1/2}$ is constant.

2. An approximate criterion for branching is that the stress at $\sim 2c$ ahead of the main crack should be able to induce persistent extension of a new $2c$ Griffith crack located there. This gives $\sigma_f c_b^{1/2} = 2(2E\gamma/\pi)^{1/2}$ in good agreement with experiment. The change in stress distribution as the crack reaches a high velocity will facilitate the extension of a Griffith crack inclined to the original crack.

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